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The hidden role of market-making in the rise of farmed salmon

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E-mail: james.robinson@lancaster.ac.uk**Keywords:** aquaculture, food systems, marketing, diets, sustainabilitySupplementary material for this article is available [online](#)

Abstract

Aquaculture is expected to play an important role in the transition towards healthy and sustainable diets, requiring dietary changes towards products with lower environmental footprints in ways that address dietary needs. Farmed Atlantic salmon accounts for 2% of global aquaculture, with productivity growth factors, such as research investment, credited with creating a high-value, globalized food commodity. However, its contribution to sustainability and nutrition goals remains unclear, and factors underlying consumer demand, such as marketing work, have been overlooked. We inspect the rise of farmed salmon in the UK, using key stakeholder interviews to understand the role of various actors (e.g. industry, retailers) in ‘making’ farmed salmon markets. Interviewees identified retailer strategies, such as labelling, and consumer-facing promotion efforts, such as health benefits, as key demand-growth factors underlying farmed salmon sales. Using data on UK seafood markets, we assess the views of interviewees, finding that farmed salmon is the most consumed aquatic food; however, it is less popular than other animal-source foods and is primarily consumed by affluent households. Our analyses discuss how market-making efforts by industry and retail actors have shaped consumer demand by promoting salmon as an accessible, nutritious and sustainable product. Accreditation and labelling have helped to develop perceptions of health and sustainability, despite farmed salmon being comparable in nutrient profile to other aquatic foods and having long-standing environmental impacts. Market-making insights could be used to enhance demand for other aquatic foods in ways that target dietary gaps by supplying affordable and sustainable products, such as mussels.

1. Introduction

Sustainable transitions towards affordable, healthy diets will require dietary shifts that disadopt unsustainable foods in favour of existing but more sustainable products (Cottrell *et al* 2021), or emerging future foods (Parodi *et al* 2018). Aquatic foods are often positioned as ways to enhance global food and nutrition security (Golden *et al* 2021, Garlock *et al* 2022) while reducing environmental footprints (Willett *et al* 2019), with farmed aquatic foods expected to play a significant role in sustainable dietary transitions (Gephart *et al* 2020). Global aquaculture produces over 100 million tons of marine algae, invertebrates and finfish annually, accounting for ~50% of aquatic foods, and will soon exceed the supply

from wild fisheries (Naylor *et al* 2021, FAO 2022). Much aquaculture supply is traded globally, with rising demand linked to rapid expansion in farming of a few marine species, such as Atlantic salmon and whiteleg shrimp (Gephart *et al* 2024). These species characterise an increasingly globalised aquatic food system, directed by a small number of corporate actors (Österblom *et al* 2015), with unclear contributions to sustainable development goals (Barton *et al* 2023) and food and nutrition security (Belton *et al* 2020).

Farmed Atlantic salmon (*Salmo salar*) is among the world’s most valuable mariculture (i.e. farmed at sea) commodities and accounts for 2% of global aquaculture production (Naylor *et al* 2021). Commercial salmon farming started in the 1960s

(Ellis *et al* 2016), expanding into a large-scale, intensive production system that now supplies over 2.7 million tons of fish globally (Pandey *et al* 2023). Multiple factors have been credited with the rise of farmed salmon as a global commodity, particularly those relating to ‘productivity growth’ (Asche *et al* 2011) such as the species’ suitability for large-scale intensive farming (e.g. fast growth), investment in technology and innovation (Regan *et al* 2021), and industry consolidation (Asche *et al* 2013). Consumer demand is often cited as a key driver of farmed salmon production, but the aspects underlying such ‘demand growth’ are less well understood (Asche *et al* 2011), particularly how market characteristics shape demand for and consumption of products. Such ‘market-making’ requires the felicitous and purposeful coalescing of multiple elements, including market actors (e.g. producers, retailers, consumers), devices (e.g. pricing mechanisms, packaging) and practices (e.g. farming, transporting, categorising) that, combined, generate a market (infra)structure that facilitates and encourages consumption (Araujo 2007, Katy Mason and Hagberg 2015). These elements are characteristic of farmed salmon retail, which has undergone substantial product categorisation (Asche *et al* 2021) and creation of new markets (Straume *et al* 2024), with the use of diverse labelling schemes to convey messages around food quality, dietary value and sustainability.

In market-making for farmed salmon, producers and retailers thus develop widely available, convenient products (e.g. salmon fillets, ready meals) and amplify characteristics that are desirable to consumers (e.g. sustainability, health properties) in order to create demand. While these actions can have positive outcomes, such as creating new markets for sustainable products, market-making has also been criticised for perpetuating false narratives. Seafood labelling schemes, for example, can promote environmentally damaging fisheries (Jacquet *et al* 2010), while marketing and lobbying for ultra-processed foods is expected to significantly raise health risks for middle-income countries (Moodie *et al* 2021). For farmed salmon, large-scale, intensive practices have long-standing sustainability issues around feed, disease and animal welfare (Kuempel *et al* 2023), while the potential for high-value mariculture products to contribute to global food and nutrition security has been questioned (Belton *et al* 2020). Even within the UK, a high-income salmon-producing country, seafood consumption remains below recommended levels (Lofstedt *et al* 2021), suggesting that farmed salmon production has not addressed dietary gaps. While the global trade (Gephart *et al* 2024) and environmental pressures (Kuempel *et al* 2023) of the salmon industry are well-documented, the effects of salmon market-making

on aquatic food markets and consumer diets remain unclear.

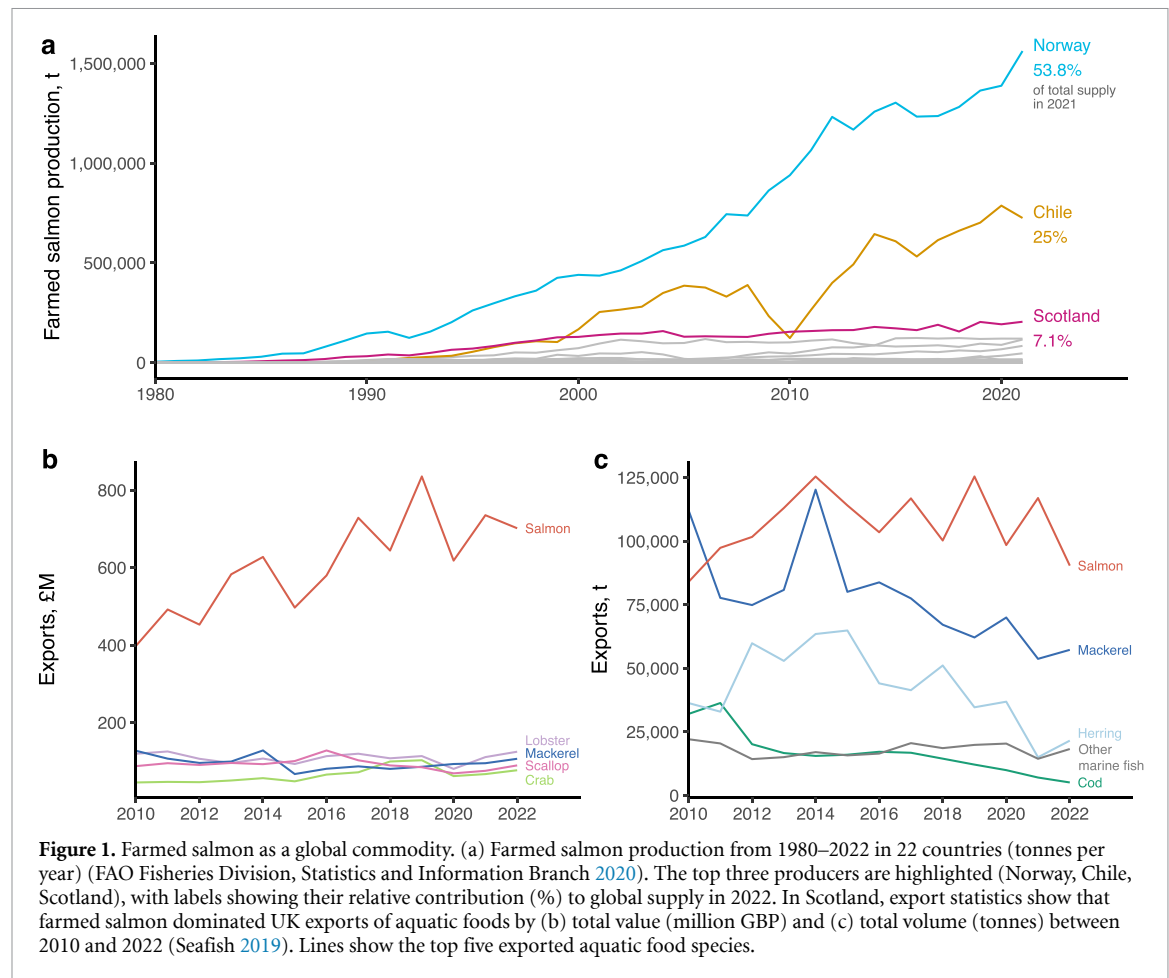
Here, we examine perspectives of farmed salmon in the UK to develop insights into market-making of aquatic foods. We interview eight expert actors involved with industry ($n = 3$), research (3), government (1) and policy advocacy (1) in Scotland to gather multi-sided views on farmed salmon, and examine productivity and demand factors underlying the rise of the salmon industry. We conduct a qualitative, thematic analysis of the interviews to identify demand-growth factors that relate to market-making and test these perspectives against data on industry supply and composition, retail and marketing, and food consumption. Finally, we discuss how aquatic food markets could be shaped to better prioritise public health and environmental sustainability, focusing on existing aquatic food markets in the UK.

2. Results

Through coding interviews, we identified ten key topics that were frequently raised by most or all of the expert actors, categorised into four themes (industry, retailers, diets, environment) (table 1). We examined interviewee perspectives on these ten topics, and confronted their statements against secondary datasets on farmed salmon production and consumption and the published literature.

2.1. Industry

All interviewees identified the significance of market consolidation and technological innovation (e.g. efficiency of feed, disease treatments) in supporting growth of the industry and the creation of a global commodity. While Norway and Chile dominate global Atlantic salmon mariculture, Scotland now accounts for 7.1% of total annual supply (205 000 t in 2021) (figure 1(a)). Supporting interviewee perspectives of a global commodity, farmed salmon production in Scotland has also increasingly oriented towards export markets. In 2022, 90 351 t of salmon was exported (£701 million, or 41% of total seafood export value), making farmed salmon the UK’s most valuable and high-volume aquatic food export (figures 1(b) and (c)). Market consolidation was perceived to have enabled ‘vertical integration’ of farmed salmon production, increasing production through economies of scale. Indeed, farmed salmon markets have consolidated over the last three decades, with the number of salmon-producing companies in the UK decreasing from 132 in 1993 to 10 in 2022, when four companies accounted for 92% of UK production (figure 2(a)). Productivity growth in the industry has been associated with technological innovations in genetic selection (e.g. faster fish growth, shorter life cycles, disease resistance) and vaccines (less mortality) and regulatory changes that permit larger and



higher numbers of cages and sites, and a higher stocking density (in freshwater systems/stages) (Ellis *et al* 2016). These technological advances and regulatory changes supported an increase from initial experimental farms in the 1970s to 203 farm sites active between 2018 and 2021 (figure 2(b)).

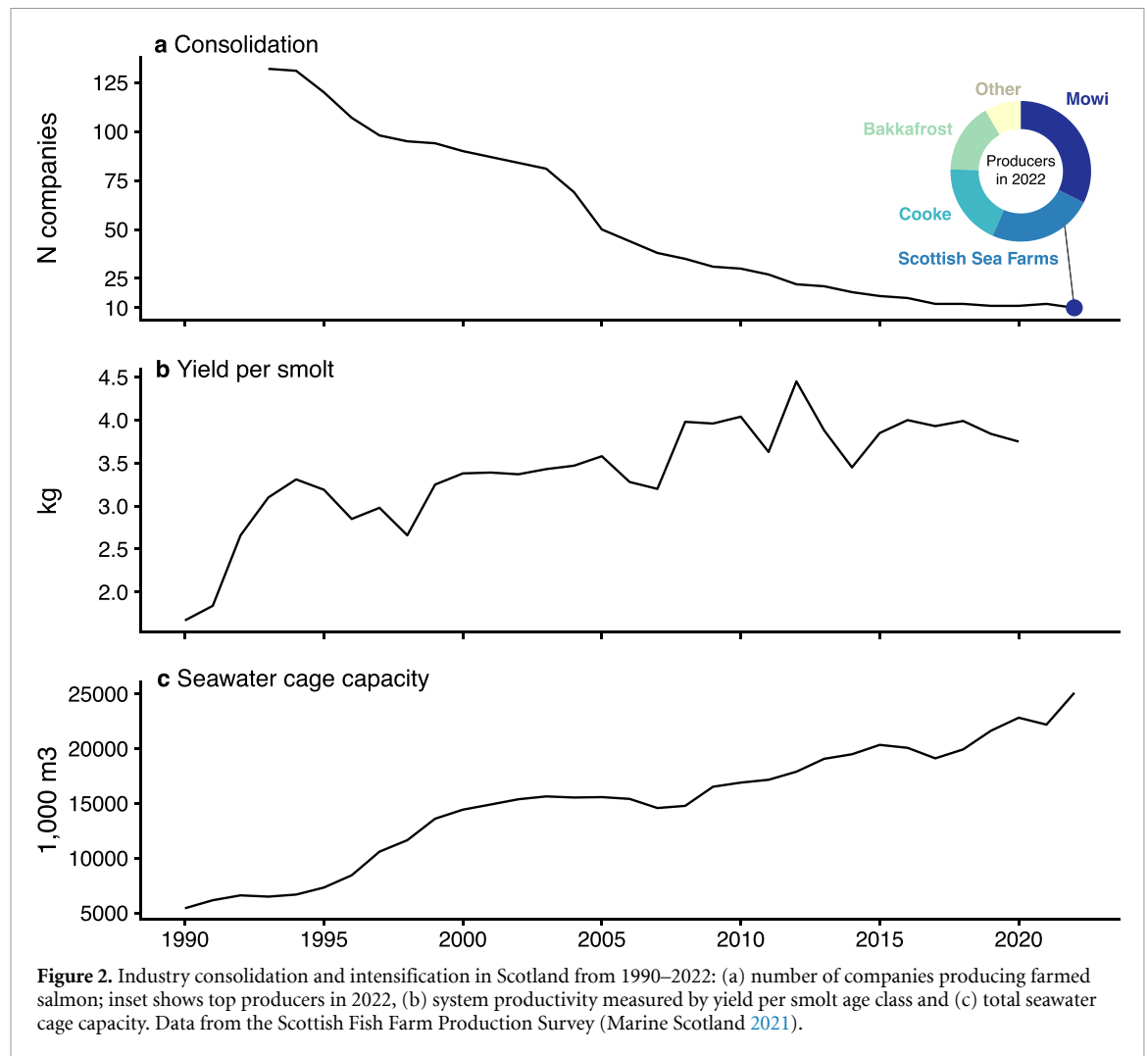
2.2. Retailers

Most interviewees viewed retailers as key players in the farmed salmon industry, and noted ‘demand-growth’ factors such as labelling and certification. The retailers mentioned in interviews were primarily supermarkets, which were associated with the growing convenience of salmon products, and the popularity of foods labelled with images associated with Scotland and healthy ecosystems (e.g. ‘*Scottish environment, pristine, great animal health and welfare*’, table S1). Accreditation labels were also associated with efforts to promote the supply of sustainable food, and are widely used by seafood producers. For example, many Scottish salmon products are associated with eco- or sustainability labels such as the Aquatic Stewardship Council, which has certified 57 farm sites in Scotland (26% of farms) (Aquaculture Stewardship Council 2024). Retail messaging was also raised by interviewees who see the health benefits of farmed salmon as important for consumers (e.g. omega-3

fatty acids, ‘*salmon is a regular health message out there*’). However, consumer demand for farmed salmon was less clearly communicated by interviewees, with farmed salmon considered both more expensive than other animal-source foods (‘*premium price*’) and ‘*one of the most purchased fish products in supermarkets*’ (table S1). Retail power was reflected in the high degree of categorisation in products, with farmed salmon considered ‘*the biggest single product within the seafood category*’, offering ‘*30–40 different products*’ to consumers.

2.3. Diets

All interviewees described the importance of consumers perceiving farmed salmon to be a convenient and healthy food, which supported retail demand (table 1). Farmed salmon was considered to be popular because it ‘*can be produced in...ways that the consumers find it easy to consume*’ (table S1). Indeed, diet surveys of UK adults show that 60% of adults consumed aquatic foods (average daily intake 40 g) between 2008 and 2018, with the ‘big five’ of salmon, tuna, cod, prawns and haddock accounting for approximately two-thirds of the average seafood diet. Steady increase in farmed salmon demand (16% to 21%) has now placed salmon as one of the top two consumed aquatic foods (for people who eat



aquatic foods), displacing haddock (−4.8%) and ‘other’ species (−6%) (figure 3(d)). Yet farmed salmon remains below other animal-source proteins. Since 1990, chicken has been the most consumed animal protein in the UK (195 g per person per week), followed by aquatic foods (135 g) and beef (86 g) (figure 3(a)). Total fish consumption has remained relatively stable since 1980, while salmon consumption steadily increased from ~2 g per week in the 1970s to 18 g by 2022. Households have increasingly consumed fish in ready meals (40% of weekly diet in 2022), and these products are substituting for long-term declines in white fish (fresh or frozen) and takeaway fish (16% of diet, combined) (figure 3(b)). While these surveys quantified consumption of fresh or frozen salmon separately from convenience forms (e.g. ready meals), our analysis shows that salmon has always been consumed less than terrestrial meats.

All interviewees identified the health benefits of salmon as being important (*‘the pinnacle of nutritional value’*), particularly noting its high omega-3 fatty acid content as a driver of consumer demand. Indeed, UK dietary guidelines recommend that adults eat two portions (2×140 g) of aquatic foods every

week, with one of those being an oily fish such as salmon (NHS 2024). A 140 g portion of farmed salmon contains over 100% of an adult’s recommended intake of omega-3 fatty acids and vitamins B12 and D, and thus can make a significant contribution to dietary nutrient intakes (Willer *et al* 2024), although at the highest retail cost of any animal-source food (figure 4(a)). However, ‘convenience’ products such as smoked salmon or fishcakes, that were also considered important in consumer demand, have ~1.5 g less eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) than fresh salmon (per 100 g) (figure 4(b)). Other oily fish such as herring, mackerel and sardine also contained comparable or higher omega-3 fatty acid levels to farmed salmon (figure 4(b)).

2.4. Environment

Most interviewees associated intensive salmon production with environmental impacts. Diverse negative sustainability perceptions were raised by all interviewees, including declines in wild salmon populations, feed supply chain issues, environmental pollution and impacts on local communities, with fish

Table 1. Stakeholder perceptions of Scottish farmed salmon. We show ten key topics that were frequently raised by interviewees. Each theme was mentioned at least once by all eight interviewees, except for fish welfare (7/8), convenience (7/8), future challenges (6/8) and retail power (5/8). Codes in bold are associated with demand-growth or market-making factors.

Theme	Interview topic	Stakeholder perspectives	Example
Industry	Global commodity	Farmed salmon is now a global seafood commodity, enabling efficiencies of scale and industry power.	<i>‘dramatic improvements in productivity and efficiency of production’</i>
	Consolidation	Salmon industry has consolidated into fewer, large companies	<i>‘there’s only really three or four producers ... in Scotland now’</i>
Retailers	Labelling and certification	Scottish brand has positive connotations for consumers	<i>‘... clear premium to be associated with the Scottish one, because it is seen as better’</i>
	Retail power	Retailers have strong influence on industry success and consumer perceptions	<i>‘salmon ... is the biggest single product within the seafood category’</i>
Diets	Health benefits	Farmed salmon contributes to healthy diets	<i>‘it is at the pinnacle of nutritional value’</i>
	Convenience	Categorisation of farmed salmon products has made it easier for consumers (e.g. ready-to-eat)	<i>‘accessible format is key’</i>
	Consumer perceptions	Accessible and popular seafood available in supermarkets	<i>‘I think it is one of the most purchased fish products in supermarkets’ ‘We can only encourage consumers to eat [what they want] to eat’</i>
Environment	Fish welfare	Scotland’s farm production is associated with good fish welfare, but also fish welfare issues in global salmon industry	<i>‘... if you ask them would turn to issues like sea lice, or use of antibiotics or polluting local waters’</i>
	Sustainability	Farmed salmon have impacts on marine ecosystems, but farmed salmon are also sustainable food choices	<i>‘a really positive story, the space of water use [...] lower carbon footprint, because I know it is got a sort of similar footprint to chicken, for example.’</i>
	Future challenges	Barriers to further growth in Scottish farmed salmon, such as marine spatial planning and climate change	<i>‘challenges that you have around disease, and particularly sea lice’</i>

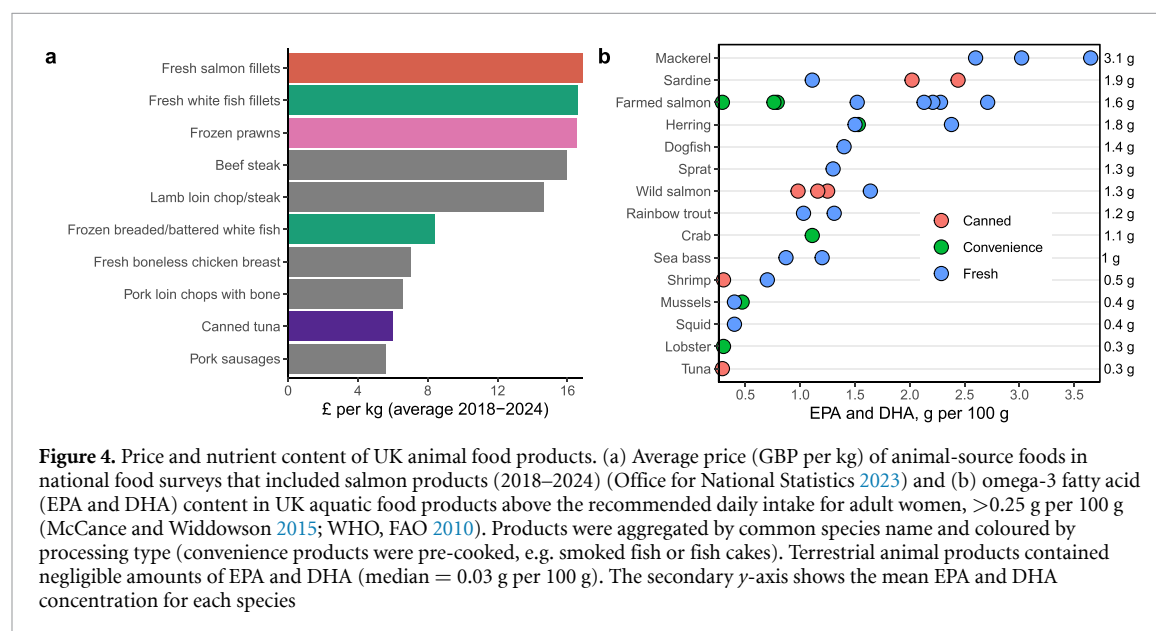
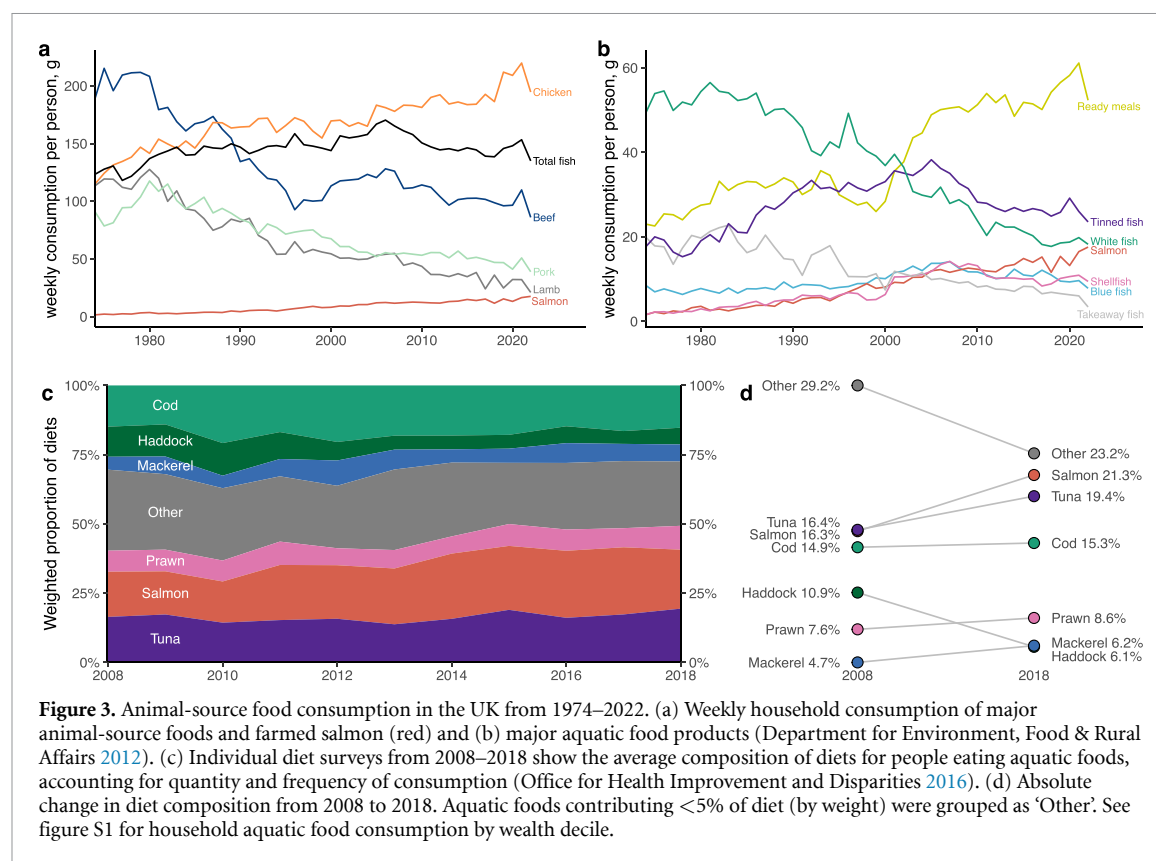
welfare identified as highly variable between Scotland and other producing countries (table 1). However, several interviewees also identified sustainability and environmental accreditation as positive attributes of farmed salmon, with ‘*farming in the sea ... intrinsically ... more sustainable than farming on land*’ (table S1). Future environmental challenges, such as climate change, were also considered to be important factors for the salmon industry. Environmental challenges were also diverse, ranging from ‘*demand for fish feed*’ to sea lice and disease, and these will continue to be key concerns (‘*the industry is very aware that sustainability debates are going to be increasingly important*’) (table S1).

3. Discussion

Favourable regulation, market consolidation and technological innovation have supported farmed salmon producers in developing a model of intensive large-scale production (Asche *et al* 2013, Ellis *et al* 2016, Graziano *et al* 2018), transforming salmon

into one of the most valuable global food commodities. One of our interviewees said ‘*we can only encourage consumers to eat [what they want] to eat*’ (table 1). However, our study suggests that the rise of demand for farmed salmon in the UK was a process that included several purposeful actions from market actors, such as the processing and creation of convenience food products, wide availability, and health and sustainability labelling, leading to positive framing of the industry from almost all interviewees. Our analyses show that consumers responded to this increase in supply by changing aquatic food preferences towards farmed salmon, positioning salmon as the most popular aquatic food product in the UK.

Market-making by industry (e.g. farmers, processors and retailers) has played a significant role in creating demand and popularity for farmed salmon. Interviewees suggested that salmon’s market appeal is linked to its accessibility, nutrient value and sustainability credentials, yet our analysis suggests that these messages can be misleading. UK food consumption datasets show that farmed salmon is less widely consumed than other animal-source proteins,



while recent increases in salmon consumption (the most popular aquatic food in 2018) have displaced other aquatic foods from diets, primarily through the popularity of convenience products. Together, long-term household consumption data and diet surveys (figure 3) suggest that increases in ready meal consumption are primarily of convenient forms of farmed salmon and tuna, consistent with evidence of growing categorisation and commodification of farmed salmon products (Asche *et al* 2021). Our

analysis thus suggests that retailers have steadily prioritised farmed salmon products over other aquatic foods, shaping seafood markets to favour more affluent consumers. With farmed salmon being among the most expensive animal-source foods (figure 4(a)), consumed primarily by affluent households (figure S1), the availability of more affordable aquatic foods may have decreased in response to salmon market-making. Indeed, in the UK between 2006 and 2018, demand for salmon increased while total seafood

consumption remained stable (figure 3(a)) and UK diets were below recommended levels of seafood intake (Harrison *et al* 2023). The rise of farmed salmon has therefore not increased aquatic food consumption in the UK, while deficiencies of nutrients concentrated in aquatic foods, such as iron and selenium, remain prevalent (Derbyshire 2018).

Food labelling can be a powerful market-making tool (Fuentes and Fuentes 2017, Cochoy 2020). Sustainability accreditation in farmed salmon products is prevalent, with all interviewees noting that retailer labelling was associated with positive perceptions of farmed salmon (e.g. 'local', 'wild' and 'organic'). Multiple accreditation schemes and ecolabels have been introduced to promote sustainability standards for aquatic foods, introducing confusion for markets and consumers, with variable success in improving sustainability (Roheim *et al* 2018). Labelling schemes used for farmed salmon have been considered as indicators of environmental or social responsibility, rather than sustainability standards (Gulbrandsen *et al* 2022). Indeed, the intensive large-scale model of salmon production has been referred to as 'precision aquaculture' (Føre *et al* 2018) and presented as an example of 'sustainable intensification' (Ellis *et al* 2016), yet all expert interviewees raised environmental and sustainability issues associated with fed-mariculture. Such a disconnect in sustainability credentials may be linked to accreditation movements that help larger companies to exert market influence by redefining sustainability (Aarset *et al* 2020) while environmental footprints remain large (e.g. emissions and land use in farmed salmon) (Newton and Little 2018, Kuempel *et al* 2023). Indeed, ASC and other aquatic food ecolabels have been criticised for failing to capture these broader systemic impacts of salmon production (Gulbrandsen *et al* 2022). Labelling thus appears to serve as a powerful demand-growth factor. The influence of industry in this area remains strong, with, for example, the UK government approving legislation in 2024 to market farmed products as 'Scottish salmon', a labelling protection previously reserved for wild-caught products (Salmon Scotland 2024).

Environmental impacts can also be diffuse and lagged, making it difficult to robustly link farm practices with environmental declines. For example, life-cycle analysis shows that over 90% of the environmental impacts of salmon are generated from fish feed, of which, in the UK, over 75% is sourced from countries distant from salmon farms (Newton and Little 2018). Positive environmental assessments of farmed salmon may also reflect recent improvements in production practices (Naylor *et al* 2021), such as decreasing dependence on marine ingredients (the use of marine oil in Norway declined from 31% to 10% from 2000 to 2016; Aas *et al* 2019), the phasing out of antibiotics (Bondad-Reantaso *et al* 2023) and

the (comparatively) low carbon footprints of net pen systems (Ziegler *et al* 2013). Nevertheless, despite having lower carbon emissions than most livestock products (Bianchi *et al* 2022, Robinson *et al* 2022), the environmental impacts of large-scale salmon farming (e.g. feed supply chains) remain substantial (Kuempel *et al* 2023). Continued supply growth in farmed salmon (Froehlich *et al* 2021) and associated growth in demand through market-making, will therefore lead to mariculture accounting for an increasing share of the environmental footprint of the UK food system, irrespective of sustainability accreditation.

The role of retail commodification in market-making can also have a strong influence on the composition of the food system. For example, consumption of 'Other' aquatic foods was partly substituted by farmed salmon (and tuna), and over one-third of salmon retail can be prepared convenience products (Straume *et al* 2024), suggesting that the contribution of salmon to intake of omega-3 fatty acids may be overstated. While such food substitutions are historically rare, aquatic food markets are highly diverse, encompassing many species, production methods and sources (Ferraro *et al* 2022). This sector may therefore be more suited to market-making efforts to reshape consumer demand. Indeed, UK consumers have already transitioned from primarily consuming wild-caught species from UK waters (cod, haddock) in the 20th century, to retail dominated by imported and farmed products (Harrison *et al* 2023). Characteristics of farmed salmon market-making could thus be used to encourage and enable the consumption of more sustainable and nutritious aquatic foods that reach people with greater dietary needs. For example, the UK also produces other farmed aquatic foods in smaller quantities (mussels, seaweed), and large quantities of wild-caught species such as herring and mackerel (Robinson *et al* 2022, Harrison *et al* 2023) destined for export markets (Graziano *et al* 2018). Reorganising the market to help increase demand for such low-impact species would contribute to the reduction of the environmental footprint from aquatic foods (Jones *et al* 2022) while addressing deficiencies in nutrients that are concentrated in aquatic foods (Willer *et al* 2021, Vogliano *et al* 2024). Such diversification (i.e. reducing the dominance of farmed salmon) would add resilience to food supply (e.g. from trade disruption) (Graziano *et al* 2018) and reduce the overall emissions of the food system (Stewart *et al* 2023). Market-making efforts towards other products could engender new markets through, for example, product categorisation (Gawel *et al* 2023). Upscaling production and demand for these markets will require further analysis of market dynamics between farmed salmon and other animal foods (terrestrial and aquatic), engagement with industry and

retailers (Ruel *et al* 2018), and investment and regulatory support (Vogliano *et al* 2024).

Dominance of individual species within seafood sectors has been historically rare (Ferraro *et al* 2022), yet the intensive production, investments in the infrastructure of processing and distribution, technology, labelling and categorisation, and other market-making practices of farmed salmon are similar to those in the poultry sector (i.e. ‘aquatic chicken’) (Gephart *et al* 2020). Our analysis suggests that demand for farmed salmon is not simply ‘naturally occurring’ or ‘intrinsic’ but has been shaped through purposeful market-making by producers and retailers, underpinned by narratives of widespread appeal, nutrient value and sustainability. We find that the dominance by farmed salmon of UK aquatic food consumption is primarily due to affluent households and, through categorisation, labelling and differentiation, may have led to reduced demand for other aquatic foods. The knock-on impacts for other aquatic food sectors remain unclear, requiring further research on how producers and retailers act to influence consumer behaviour. Uncovering market-making in food systems can reveal ways of promoting the consumption of new or alternative products, providing means to address consumer barriers for more affordable, nutritious and sustainable aquatic foods (Gawel *et al* 2023).

4. Methods

Our ‘marketography’ of UK farmed salmon used ‘interviews, observation, documentary work, [and] historical record’ to trace the market-making efforts of actors across space and time (Roscoe and Loza 2019).

4.1. Interviews with expert stakeholders

Initial participants were recruited following identification of the main players in the market through publicly available data and were contacted via online channels. From this, a process of snowballing (Thorpe *et al* 2008) allowed for further potential participants to be selected. Sampling through snowballing poses the risk of distorting bias (Hammersley and Atkinson 2019) (e.g. only meeting colleagues); however, the multi-sided nature of our approach to recruitment ensured that we accounted for diverse perspectives. We contacted a variety of actors, recruiting a final count of eight people for interviews, including activists, aquaculture researchers, industry leaders and policymakers who were identifiably involved with salmon farming (e.g. had published research on salmon aquaculture). This number of interviews is appropriate for qualitative data analysis, which prioritises depth over breadth of data, and adopts thematic saturation (the point at which no new themes are emerging) as a key determinant of sample size

(Easterby-Smith *et al* 2015). One of the authors also attended one of the largest industry events in Scotland in 2023 to network and identify potential participants; this resulted in one of the expert interviews.

Interviews followed a semi-structured guide which allowed adaptation and exploration during conversations (see supplementary methods). Thus, the authors made decisions about ‘which line of questioning [...] to explore further’ (Thorpe *et al* 2008) depending on which points were raised by the interviewees. Interviews were conducted either in person in Scotland or online, and both authors were present at all but two interviews. In this case, the non-present co-author listened to a recording of the interview. All interviews, in person and online, were recorded and transcribed by an independent transcriber and subsequently cross-checked by the authors. Transcriptions formed the dataset of the thematic analysis to identify key topics raised by the participants. Analysis was performed through open, axial and selective coding in which every ‘pass-through’ involved identifying and refining keywords and grouping into topics (Neuman 2012). Both authors conducted an independent analysis of the texts to identify key topics, which was followed by a process of contrasting, discussion and agreement.

4.2. Datasets on UK farmed salmon and aquatic foods

We extracted data on the global production of farmed salmon (in tonnes) from the FAO Fishery and Aquaculture Statistics Databases (FAO Fisheries Division, Statistics and Information Branch 2020), and examined the relative contributions of the top three producing countries (Norway, Chile, Scotland) from 1980 to 2021. We gathered additional data on Scotland’s farmed salmon production, trade and industry. We extracted the trade quantity and value of major aquatic foods from 2010–2022, using EUMOFA data provided by Seafish (Seafish 2019). From the Scottish Fish Farm Production Survey (Marine Scotland 2021), we extracted data on industry composition (the number of companies from 1993–2022) and farm system productivity (the number of sites, smolt size and cage capacity from 1990–2022).

We also used public food consumption databases to understand the contribution of farmed salmon to UK diets relative to other terrestrial and aquatic animal-source foods. We assessed demand for animal-source foods using family food datasets collected by the UK government (Department for Environment, Food & Rural Affairs 2012). These datasets estimate the average quantity of food products purchased per person per week, from 1974 to 2022, providing a standardised metric of

long-term demand for major UK animal-source foods. We extracted all animal-source foods recorded in this dataset and further categorised aquatic foods into groups of similar species (e.g. blue fish, white fish). We complemented these household consumption estimates with data from individual diet recall surveys, collected by the National Diet and Nutrition Survey (Office for Health Improvement and Disparities 2016). These surveys allowed us to understand the relative consumption of aquatic food products at greater resolution than the household consumption dataset, containing information on portion sizes and consumption frequency of 43 aquatic food products, from 2008–2018. We examined annual changes in aquatic food consumption by adults, aggregating products to common species names (e.g. salmon, cod, tuna) and estimating the relative proportion (i.e. frequency $\times$ portion size) of each product, for individuals who consumed aquatic foods. We visualized all species that comprised $\geq 5\%$ of diets (cod, haddock, prawn, salmon, tuna, mackerel) and grouped the remaining species as ‘Other’.

Finally, we examined aspects of market-making in farmed salmon using data on the average price of animal-source foods, and omega-3 fatty acid content in aquatic foods. Price estimates were extracted from the Office for National Statistics dataset on inflation-corrected prices of commonly consumed foods by UK households (Office for National Statistics 2023). This dataset tracks the monthly price of a food basket of popular items, corrected for inflation, including farmed salmon, four other aquatic foods and five terrestrial animal-source foods. We estimated the average price per kg of these foods from 2018–2023. We next extracted omega-3 fatty acid estimates from McCance and Widdowson (2015) and identified all foods with EPA and DHA concentrations above recommended dietary intakes for adult women (>250 mg per 100 g) (WHO, FAO 2010).

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://github.com/jpwrobinson/salmon-market>.

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Ethical statement

This research was conducted in accordance with the principles embodied in the Declaration of Helsinki and in accordance with local statutory requirements. The study was approved by the Faculty of Arts and Social Sciences and Lancaster University Management School Research Ethics Committee (FASLUMS-2023-3402-RECR-1). All interviewees were provided a Participant Information Sheet, which detailed the aims of the research, and gave written informed consent to participate in the study. All participant data has been anonymized and stored in encrypted folders, according to Lancaster University’s ethics regulations.

Author contributions

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 Conceptualization (equal), Data curation (equal), Formal analysis (equal), Funding acquisition (equal), Investigation (equal), Methodology (equal), Project administration (equal), Visualization (equal), Writing – original draft (equal), Writing – review & editing (equal)

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